

Math Functions

Rakib Mahmud

abs() Function in C

- In the C programming language, the use of the function abs is to return the absolute value of an integer.
- By absolute value, it means the function returns the positive value of an integer.
- Let's say we have an int value -3, and we want to know what the absolute value of -3 is. So, we will use the absolute function and return the absolute value of -3, which is 3.

Program

main.c			Run	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 4 int main () { 5 int x, y; 6 7 x = abs(5); 8 printf("absolute value of x = %d\n", x); 9 10 y = abs(-10); 11 printf("absolute value of y = %d\n", y); 12 13 return(0); 14 } 15</pre>	<pre>/tmp/Y0ouCaEU8C.o absolute value of x = 5 absolute value of y = 10 === Code Execution Successful ===</pre>			

sqrt() in C

- The sqrt() function in C returns the square root of the number.
- It is defined in the math.h header file.
- The sqrt() function takes a single number(in double) as the input.
- The sqrt() function returns a double number value as the output.
- However, if the input is negative, it returns -nan. To understand why that happens, see Exceptions.

Syntax: `double sqrt(double x);`

Program

main.c	  Run	Output
<pre>1 #include<stdio.h> 2 #include<math.h> 3 4 int main() 5 { 6 int a = -9; 7 printf("The square root of %d: %.4lf\n",a, sqrt(a)); 8 return 0; 9 } 10</pre>	<pre>/tmp/18dpWYHCFH.o The square root of -9: -nan === Code Execution Successful ===</pre>	

Program

main.c			Run	Output
<pre>1 #include<stdio.h> 2 #include<math.h> 3 4 int main() 5 { 6 int a =9; 7 printf("The square root of %d: %.4lf\n",a, sqrt(a)); 8 return 0; 9 } 10</pre>	<pre>/tmp/GtyEf6fb0Z.o The square root of 9: 3.0000 === Code Execution Successful ===</pre>			

Power Using pow() Function

- The pow() function is used to calculate the power of a number in C.
- It takes double as input and returns double as output.
- The pow() function takes two arguments x and y.
 - x: The variable x represents the base value, whose power needs to be calculated. It is of the type double.
 - y: The variable y represents the exponent value. It is of the type double.

Syntax: `double pow (double x, double y);`

Program using pow() Function

main.c	Output
<pre>1 #include <math.h> 2 #include <stdio.h> 3 4 int main() 5 { 6 double x = 6.1, y = 4.8; 7 8 // Storing the answer in result. 9 double result = pow(x, y); 10 printf("%.2lf", result); 11 12 return 0; 13 } 14 </pre>	<pre>/tmp/1heykb6ITU.o 5882.79 === Code Execution Successful ===</pre>

ceil() in C

- This C library function returns the smallest integer value greater than or equal to x.

Syntax: `double ceil(double x);`

Program

main.c			Run	Output
<pre>1 #include <math.h> 2 #include <stdio.h> 3 4 int main() 5 { 6 float val1, val2, val3, val4; 7 8 val1 = 1.6; 9 val2 = 1.2; 10 val3 = -2.8; 11 val4 = -2.3; 12 13 printf("value1 = %.1lf\n", ceil(val1)); 14 printf("value2 = %.1lf\n", ceil(val2)); 15 printf("value3 = %.1lf\n", ceil(val3)); 16 printf("value4 = %.1lf\n", ceil(val4)); 17 18 return (0); 19 }</pre>				<pre>/tmp/8eTaFn5pPa.o value1 = 2.0 value2 = 2.0 value3 = -2.0 value4 = -2.0 === Code Execution Successful ===</pre>

floor() in C

- The C library function `double floor(double x)` returns the largest integer value less than or equal to `x`.

Syntax: `double floor(double x);`

trunc() in C language

- This function is used to remove digits after decimal point and return the modified decimal number.

Syntax: `double trunc(double x)`

Program

main.c	  	Output
<pre>1 #include <stdio.h> 2 #include <math.h> 3 4 // Driver function 5 int main() 6 { 7 // using trunc function which return 8 // Truncated value of the input 9 double x1 = 2.0, x2 = 3.9, x3 = -3.3, x4 = 4.9; 10 printf(" Truncated value is %lf \n", trunc(x1)); 11 printf(" Truncated value is %lf \n", trunc(x2)); 12 13 // For negative values 14 printf(" Truncated value is %lf \n", trunc(x3)); 15 printf(" Truncated value is %lf \n", trunc(x4)); 16 return 0; 17 }</pre>	<pre>/tmp/Sj1DKu044s.o Truncated value is 2.000000 Truncated value is 3.000000 Truncated value is -3.000000 Truncated value is 4.000000 === Code Execution Successful ===</pre>	

round() Function in C

- C round() is a built-in library function that rounds a floating-point number to the nearest integer.
- If the fractional part of the number is 0.5 or greater, the argument is rounded away from zero.
- If the fractional part is less than 0.5, the argument is rounded towards zero.

Syntax: `double round(double x);`

Program

main.c

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3 #include<math.h>
4 int main() {
5
6     double num1=2.5;
7     double num2=3.8;
8     double num3=-2.5;
9     double num4=-3.5;
10    printf("Rounded value of num1 : %.1lf\n",round(num1));
11    printf("Rounded value of num2 : %.1lf\n",round(num2));
12    printf("Rounded value of num3 : %.1lf\n",round(num3));
13    printf("Rounded value of num4 : %.1lf\n",round(num4));
14 }
```

Output

/tmp/3TZejVZrko.o

Rounded value of num1 : 3.0

Rounded value of num2 : 4.0

Rounded value of num3 : -3.0

Rounded value of num4 : -4.0

=== Code Execution Successful ===

Some other functions

Math Notation	Function Signature	Description
$\ln(x)$	<code>double log(double x)</code>	a function that evaluates the natural logarithm of x
$\text{Log}_{10}(x)$	<code>double log10 (double x)</code>	a function that evaluates the logarithm of x in base 10
e^x	<code>double exp(double x)</code>	a function that evaluates $e(2.718282)$ to the power x
$\cos(x)$	<code>double cos(double x)</code>	a function that evaluates cosine of x

Program

main.c	  	Output
<pre>1 #include <math.h> 2 #include <stdio.h> 3 4 int main() 5 { 6 double x = 6.1, y = 4.8; 7 8 // Storing the answer in result. 9 double result1 = log(x); 10 printf("ln value is : %.2lf\n", result1); 11 12 double result2 = log10(x); 13 printf("Log10 value is : %.2lf\n", result2); 14 15 printf("Exponential value is : %lf\n",exp(0)); 16 17 printf("Cosine value is : %lf\n",cos(90)); 18 return 0; 19 } 20</pre>	<pre>/tmp/Ys81v3Ajss.o ln value is : 1.81 Log10 value is : 0.79 Exponential value is : 1.000000 Cosine value is : -0.448074 === Code Execution Successful ===</pre>	

Lab work

- Consider these numbers

num1=-3.5, num2= 5.3, num3= 6.1, num4= 7.1

- Perform the following operations step by step:
 - Round the value of num1 and put it in num1; also print it.
 - Calculate the square root of num2 and put it in num2 ; also print it.
 - Determine the floor value of num3 and put it in num3 ; also print it.
 - Determine the ceil value of num4 and put it in num4 ; also print it.
 - Then do $X = (\text{num3})^{(\text{num2})}$, $Y = \text{num1} + \text{num4} + X$; also print it.
 - Take the absolute value of Y, put it into the Y, also print it.
 - Then apply natural logarithm in Y and show the results.

Lab work

- Final result: 4.187594